How you implement each of these components in your agentic systems may, in part, be guided by the architecture of your chosen agentic platform. Now that you understand the nuances of each component, you also have the knowledge to guide you in selecting the right agent system that fits your application and business use case. Regardless of your application, you'll want to employ several agentic components in almost all cases.

As agentic systems mature and LLMs themselves get smarter, some of the components we today consider external may be closely integrated. We've already seen reasoning and planning be integrated into a model such as Strawberry. Certainly, as we approach the theoretical artificial general intelligence milestone, we may see models capable of performing long-term self-evaluation and feedback.

In any case, I hope you enjoyed this journey with me into this incredible frontier of a new and emerging technology that will certainly alter our perception of work and how we undertake it through agents.

11.6 Exercises

Use the following exercises to improve your knowledge of the material:

- *Exercise 1*—Implement a Simple Planning Agent (Beginner)

Objective —Learn how to implement a basic planning agent using a prompt to generate a sequence of actions.

Tasks:

  - Create an agent that receives a goal, breaks it into steps, and executes those steps sequentially.
  - Define a simple goal, such as retrieving information from Wikipedia and saving it to a file.
  - Implement the agent using a basic planner prompt (refer to the planner example in section 11.3).
  - Run the agent, and evaluate how well it plans and executes each step.